INDIAN INSTITUTE OF TECHNOLOGY ROORKEE

DEPARTMENT OF ELECTRONICS AND COMMUNICATION ENGINEERING

DIGITAL COMMUNICATION

Quiz-2, Spring Semester, 2025

Time Allowed: ONE hour **Total Marks: 15**

1. A voice signal in the range 300 to 3300 Hz is sampled at 8000 samples/sec. We may transmit these samples directly as PAM pulses or we may first convert each sample to a pulse-code modulation (PCM) format and use binary PCM waveforms for transmission.

   (a) What is the minimum system bandwidth required for the detection of PAM with no ISI and with a filter roll-off characteristic of $\alpha = N \times 10^{-4}$ where N represents the last four digits of your enrollment number? (1 mark)

   (b) Using the same filter roll-off characteristic, what is the minimum bandwidth required for the detection of binary PCM waveforms if the samples are quantized to eight levels? (1 mark)

Voice signal range is 300 Hz to 3300 Hz

Sampling frequency $f_s = 8000$ samples/sec $= R_s$

(a) System bandwidth $B_T = W(1+\alpha)$

For PAM, $W = \frac{R_s}{2} = \frac{8000}{2} = 4000$

$B_T = 4000(1+\alpha) = 4000(1+0.0001N)$

(b) PCM using 8-level quantization

Bit rate $= R_b = 8000 \times \log_2 8 = 24000$ bits/sec

$B_T = \frac{R_b}{2}(1+\alpha) = 12000(1+0.0001N)$

(Refer to Q1 of Tutorial-4)

2. For an AWGN channel, prove that the outputs of correlators in a correlator receiver are statistically independent of each other. (3 marks)

* Derived in the lecture class

* Pages 319 – 320 of Reference Book i.e. "Communication Systems" by Simon Haykin, 4th edition

3. Assume that N represents the last four digits of your enrollment number. A binary communication system transmits bits, x_n, where $x_n \in \{-1, +1\}$ with equal probability. The overall pulse response $p(t)$ is

$$p_k = p(kT) = \begin{cases} 1, & \text{k} = 0 \\ N \times 10^{-4}, & \text{k} = 1, 2 \\ 0, & \text{elsewhere} \end{cases}$$

(a) Determine the possible values for the ISI and their probabilities. (2 marks)

(b) Derive an expression for the probability of error. (2 marks)

The receiver output samples are

$$y_i = \underbrace{x_i p(0)}_{\substack{\text{desired} \\ \text{component}}} + \underbrace{\sum_{j \neq i} x_j \, p[(i-j)T_b]}_{\substack{\text{ISI} \\ \text{component}}} + \underbrace{n_i}_{\substack{\text{noise} \\ \text{component}}}$$

In this case, $y_i = x_i p(0) + \underbrace{x_{i-1} p(T_b) + x_{i-2} p(2T_b)}_{\text{ISI}} + n_i$

$\Rightarrow y_i = x_i + 0.0001N\,(x_{i-1} + x_{i-2}) + n_i$

(a)

x_{i-1}	x_{i-2}	ISI	Probability
-1	-1	$-0.0002N$	$\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$
-1	$+1$	0	$\frac{1}{4}$
$+1$	-1	0	$\frac{1}{4}$
$+1$	$+1$	$0.0002N$	$1/4$

Therefore, $\text{ISI} = \begin{cases} -0.0002N & \text{with prob. } 1/4 \\ 0 & \text{with prob. } 1/2 \\ 0.0002N & \text{with prob. } 1/4 \end{cases}$

If $x_i = +1$, then $y_i = 1 + \text{ISI} + n_i$

(b) Let us assume that variance of n_i is σ_n^2. Then, the prob. of error $P_e = \frac{1}{4} Q\left(\frac{1-0.0002N}{\sigma_n}\right) + \frac{1}{2} Q\left(\frac{1}{\sigma_n}\right) + \frac{1}{4} Q\left(\frac{1+0.0002N}{\sigma_n}\right)$

(Refer to Q2 of Tutorial 5) * You can also write in terms of erfc() if it is convenient to you.

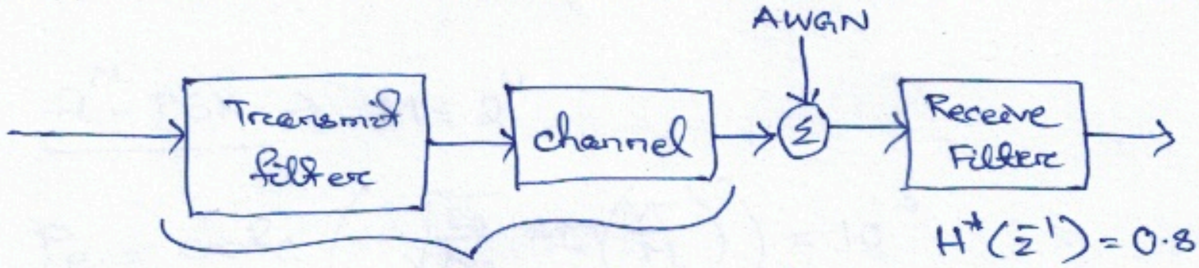
4. Consider the transfer function for the combined system of transmit filter and channel as

$$H(z) = 0.8 + \left(N \times 10^{-4}\right) z^{-1}$$

where N represents the last four digits of your enrollment number. Assuming that the one-sided noise power spectral density is 0.1 W/Hz.

(a) Determine the tap coefficients of a three-tap zero-forcing equalizer. (1 mark)

(b) Determine the tap coefficients of a three-tap MMSE based equalizer. (2 marks)



$$H^*(z^{-1}) = 0.8 + (0.0001N)z$$

$$H(z) = 0.8 + (0.0001N)z^{-1}$$

$$X(z) = H(z)\,H^*(z^{-1}) = \left[0.8 + (0.0001N)z^{-1}\right]\left[0.8 + (0.0001N)z\right]$$

$$= 0.64 + 0.0001N\left(z^{-1} + z\right) + (0.0001N)^2$$

$$= 0.64 + (0.0001N)^2 + 0.0001N\left(z^{-1} + z\right)$$

Let $N = 6100 \Rightarrow X(z) = 0.64 + (0.61)^2 + 0.61\left(z^{-1} + z\right)$

$$= 1.0121 + 0.61\left(z^{-1} + z\right)$$

(a)
$$\begin{bmatrix} 1.0121 & 0.61 & 0 \\ 0.61 & 1.0121 & 0.61 \\ 0 & 0.61 & 1.0121 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} -2.1774 \\ 3.6128 \\ -2.1774 \end{bmatrix}$$

(Refer to Q4 of Tutorial-4)

(b) One-sided noise PSD $= 0.1$ W/Hz

Two-sided " " $= \frac{0.1}{2} = 0.05$ W/Hz

$$\begin{bmatrix} 1.0121+0.05 & 0.61 & 0 \\ 0.61 & 1.0121+0.05 & 0.61 \\ 0 & 0.61 & 1.0121+0.05 \end{bmatrix} \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} 0.61 \\ 0.8 \\ 0 \end{bmatrix} \Rightarrow \begin{bmatrix} w_1 \\ w_2 \\ w_3 \end{bmatrix} = \begin{bmatrix} -0.1402 \\ 1.2442 \\ -0.7146 \end{bmatrix}$$

5. Among two bandpass data transmission systems, one system uses 2^N-PSK, and the other system uses 2^N-QAM where N represents the last four digits of your enrollment number. The required average probability of symbol error for both systems is equal to 10^{-3}.

(a) Determine E_s/N_0 in terms of erfc() required for the PSK system. (1 mark)

(b) Determine the average E_s/N_0 in terms of erfc() required for QAM system. (1 mark)

(c) Compare the energy-to-noise PSD ratio requirements of these two systems. (1 mark)

(a) 2^N-PSK $\rightarrow$ $M = 2^N$

$$P_e = \text{erfc}\left(\sqrt{\frac{E_s}{N_0}}\sin\left(\frac{\pi}{M}\right)\right) = 10^{-3}$$

$$\Rightarrow \frac{E_s}{N_0} = \left(\frac{\text{erfc}^{-1}(10^{-3})}{\sin(\pi/M)}\right)^2$$

(b) 2^N-QAM $\rightarrow$

$$P_e = 2\left(1 - \frac{1}{\sqrt{M}}\right)\text{erfc}\left(\sqrt{\frac{3E_{av}}{2(M-1)N_0}}\right) = 10^{-3}$$

$$\Rightarrow \frac{E_{av}}{N_0} = \left[\text{erfc}^{-1}\left(\frac{10^{-3}}{2\left(1 - \frac{1}{\sqrt{M}}\right)}\right)\right]^2 \times \frac{2(M-1)}{3}$$

(c) 2^N-PSK demands more symbol energy than 2^N-QAM for the same P_e.

Problem 6.16 of Reference Book i.e. "Communication Systems" by Simon Haykin, 4th edition